Project 3: Aircraft Classification with a Pretrained ResNet-18

Group 4

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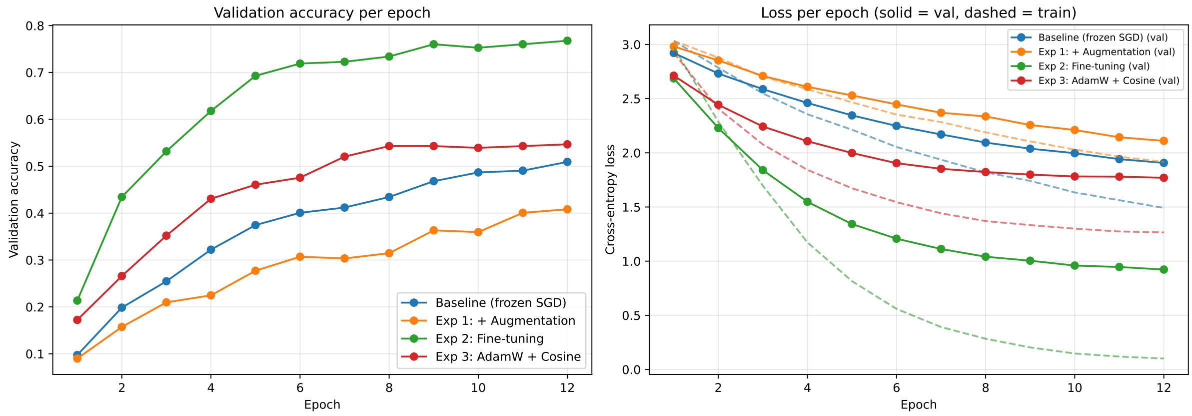

Figure 1. Validation accuracy (left) and cross-entropy loss (right) per epoch for the baseline and all three experiments, on shared axes. Dashed lines on the right are training loss. Fine-tuning (green) separates from the three frozen-backbone runs within two epochs.

A pretrained ResNet-18 (ImageNet weights) was adapted by transfer learning to classify the 20-class FGVC-Aircraft subset. The `trainval` folder of 1,331 images was divided by a stratified 1,064/267 split into training and validation sets; the 669 test images were held out and used once, for the final evaluation only. Every image was resized to 224×224 and normalised with the ImageNet mean and standard deviation to match the pretrained backbone, and the 1000-way ImageNet head was replaced with a fresh 20-way linear layer. The three experiments follow a controlled design: each alters a single variable relative to a common baseline and trains for 12 epochs, so any change in validation accuracy is attributable to that one factor.

1. Baseline model

The baseline uses ResNet-18 as a fixed feature extractor: the convolutional backbone is frozen and only the new classification head is trained, with plain SGD (learning rate 0.001, momentum 0.9) on un-augmented images. This establishes the performance floor every experiment must beat. It reached a best validation accuracy of **0.5094** (Table 1; blue curve, Figure 1). The frozen ImageNet features separate the twenty aircraft types moderately well, but because the backbone cannot adapt, the head alone cannot capture the fine-grained differences between visually similar variants.

2. Experiment 1: Data augmentation

Hypothesis: randomly flipping, rotating ($\pm 15°$) and colour-jittering the training images exposes the frozen extractor to more varied views, reducing over-fitting and improving validation accuracy. **Result: not supported.** Augmentation lowered best validation accuracy to 0.4082, a drop of -0.1011 below the baseline (orange curve, Figure 1). With the backbone frozen the network cannot adapt its features to the distorted inputs, so augmentation pushes images into regions of the fixed ImageNet feature space that the linear head has not learned, and 12 epochs are too few to compensate. Augmentation therefore hurts a pure feature-extraction setup on this dataset.

3. Experiment 2: Fine-tuning the backbone

Hypothesis: ImageNet features are generic, so unfreezing the backbone and letting every layer adapt to aircraft should capture fine-grained variant differences and lift accuracy well above the frozen baseline. **Result: strongly supported.** Unfreezing the backbone was the only change from the baseline, keeping the identical SGD optimiser and 0.001 learning rate; it raised best validation accuracy to **0.7678**, a gain of $+0.2584$ (green curve, Figure 1). This is by far the largest single improvement, confirming that adapting the convolutional features, and not just the classifier, is what fine-grained aircraft recognition requires.

4. Experiment 3: AdamW and cosine annealing

Hypothesis: replacing plain SGD with AdamW (adaptive updates with decoupled weight decay) and decaying the learning rate on a cosine schedule should give faster, more stable convergence of the frozen head and a modest accuracy gain. **Result: supported, modestly.** With the backbone frozen and images un-augmented, best validation accuracy rose to 0.5468, a gain of $+0.0375$ over the baseline (red curve, Figure 1). AdamW converges faster and to a slightly better optimum than SGD, but with the backbone still frozen the ceiling is low, so the gain is real but small beside fine-tuning.

Optimiser, decoupled weight decay and schedule are varied together as one variable: the optimisation strategy, the axis the task description names. They are not separable the way a single hyperparameter is: decoupled weight decay is the very thing that distinguishes AdamW from Adam, and AdamW under cosine annealing is a different procedure rather than another setting of the same one. Everything outside that axis (frozen backbone, un-augmented data, 12 epochs, batch size, 0.001 base learning rate) is identical to the baseline, so the comparison stays controlled.

5. Best model

Rationale. The final model keeps only the changes that improved validation accuracy over the baseline. Fine-tuning ($+0.2584$) and AdamW with cosine annealing

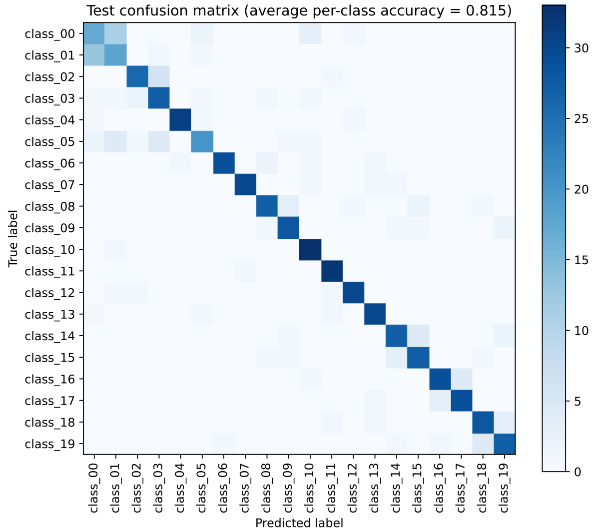

Figure 2: Confusion matrix of the final model on the 669 test images, shown as a heatmap; rows are true classes, columns predicted. The off-diagonal counts quoted in the text are printed by `main_report.ipynb` rather than read from this figure.

Model	Best val. acc.	Δ base
Baseline (frozen, SGD)	0.5094	n/a
Exp 1: + Augmentation	0.4082	−0.1011
Exp 2: Fine-tuning	0.7678	+0.2584
Exp 3: AdamW + cosine	0.5468	+0.0375

Table 1: Best validation accuracy of the baseline and each single-variable experiment.

(+0.0375) both helped and are retained; augmentation is excluded, because it reduced accuracy and so is not a winning ingredient. The final model is therefore an unfrozen ResNet-18 trained with AdamW and cosine annealing on un-augmented images. It was chosen on best mean validation accuracy rather than on stability or hardest-class performance, because the margin between fine-tuning and every alternative (+0.2584) is far larger than the epoch-to-epoch variation visible in Figure 1.

Combining two separately validated changes alters the optimisation problem, so the learning rate was cut to 0.0001 for the final run: AdamW's adaptive steps on every backbone layer at 0.001 are more aggressive than either experiment tested, since Experiment 2 used SGD on all layers and Experiment 3 used AdamW on the head alone. This is the one deliberate departure from a validated configuration, and it is a design judgement rather than a measured result: once the model is refit on all of `trainval`, no validation data survives on which to tune it.

Result. The final model was retrained on the entire `trainval` set (all 1,331 images, training plus validation, no validation split) for 15 epochs, and every figure reported here is measured on the 669 held-out test images with that refit model, evaluated once. It achieves an average per-class accuracy of **0.8148**, comfortably exceeding the required 0.75, with overall accuracy 0.8146 and a macro-averaged F1 of 0.8152. Average per-class accuracy, the mean of the 20 per-class recall values, is the reported metric rather than overall accuracy because it weights every aircraft type equally instead of letting the larger or easier classes dominate. The two agree closely here only because the test split is balanced. Macro F1 is given alongside because recall alone cannot see a class the model over-predicts. The test confusion matrix is Figure 2.

6. Potential and limitations

The strong diagonal in Figure 2 shows most classes are recognised reliably: 7 of 20 score above 0.87 and the best (class_10, class_11) reach 0.971 and 0.970. The clear weakness is class_00 at 0.500, which is confused with class_01 in both directions: 13 class_01 images are predicted as class_00 and 11 the other way, indicating two visually near-identical variants the model cannot separate. class_05 (0.606) and class_02 (0.788) fail similarly against look-alike types. The fine-tuned model also reaches 1.00 training accuracy within a few epochs, so the gap of roughly 0.19 to test performance reflects real overfitting, which the small per-class support makes hard to avoid: at roughly 669/20 test images per class, a single extra error moves a class score by about three points.

For the target application the method is promising: 0.8148 average per-class accuracy over twenty types, from barely a thousand labelled images and a backbone that trains in minutes, shows transfer learning suits a catalogue task with narrow visual differences. The anticipated limitations are the near-identical variants above, unlikely to separate without finer-grained supervision or higher-resolution inputs; the small per-class support, which makes every class score coarse; and uncalibrated output scores, which should not be read as probabilities in deployment. The obvious next steps are reintroducing light augmentation now the backbone is trainable (augmentation regularises a fine-tuned network differently from a frozen one, so Experiment 1 does not settle the question), tuning weight decay or dropout, early stopping, and a deeper backbone such as ResNet-34 for the hardest look-alike pairs.